

Caffeine as an energy storage material for next-generation lithium batteries

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ABSTRACT

Organic compounds are attracting attention as potential candidates for next-generation energy storage materials for rechargeable batteries. The utilization of redox centers in naturally occurring and human-edible organic compounds has excellent potential for designing sustainable and safe energy storage materials. However, an in-depth understanding of the energy storage mechanism is an essential prerequisite for its use and optimization as an electrode material. In this study, we applied caffeine as an electrode material in lithium batteries and revealed the energy storage mechanism for the first time. Two equivalents of electrons and lithium-ions participate in redox reactions during the charge-discharge process, providing a reversible capacity of 265 mAh g⁻¹ in a voltage window of 1.5–4.3 V. Furthermore, it was discovered that a new redox reaction site can be controlled through molecular tuning by halogenation, and the newly created redox site shows that oxidation voltage increases in proportion to the electronegativity of halogen elements in the order of I, Br, Cl, and F. These results demonstrate the applicability of nature-derived organic materials as energy storage materials and a feasible strategy to design tunable redox-active sites and their redox potential through molecular-scale engineering.

1. Introduction

The recent climate change phenomenon has further enhanced interest in environmental issues. Accordingly, eco-friendly energy storage and conversion system are becoming increasingly important. The development of high-performance rechargeable batteries has become significantly promising as a solution [1,2]. Lithium-ion batteries (LIBs) are currently one of the dominant technologies in energy storage systems [3]. Structurally, the main components of LIBs are cathodes, anodes, separators, electrolytes, etc. Among these constituents, the advancement of the cathode has been regarded as a main factor in determining the overall performance improvement of LIBs [4–6]. Currently, the leading materials in the cathodes are transition metal-based inorganic compounds. The most representative materials are LiCoO₂ and its derivatives, in which Co ions are substituted with Mn, Ni, and other metal ions. Moreover, the strategy of developing these cathode materials for the small devices market was successfully applied to medium-to-large markets, such as electric vehicles and energy storage systems [7,8]. As the application of transition metal-based inorganic compounds was extended further, there are growing concerns that the

finite reserves of these materials will create further challenges such as high cost and sustainability [9,10]. An organic-based electrode system is emerging as a promising alternative to solve these problems. Unlike transition metal-based electrode materials, organic-based electrode materials largely consists H, C, N, O, and S, all of which occur abundantly in nature [11]. In addition, low toxicity, low carbon footprint, and sustainability are general characteristics of organic materials, which are attractive points for the application of organic materials as energy storage materials [12]. Utilizing these abundant and inexpensive resources is an appealing strategy to design a variety of electrode materials while simultaneously addressing issues in transition metal-based electrode materials [13,14].

Concerning organic-based electrode materials, there are several types of redox-active energy storage materials such as conductive polymers, carbonyl, organosulfur compounds, and organic radicals [15–25]. These organic-based electrode materials can be artificially synthesized and is also found in nature. The utilization of nature-derived materials with low toxicity is an appealing approach concerning eco-friendliness, sustainability, and biological safety. Over the past decade, various natural and bioderived (or inspired) organic materials

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have been reported and evaluated for electrode materials. Some of them have already demonstrated superior performance over conventional inorganic materials [26–30]. However, organic-based electrode materials have critical issues that need to be addressed before practical use, such as low electrical conductivity, low redox potential, and high solubility in liquid organic electrolyte systems [31]. To overcome these

intrinsic challenges and further develop the organic-based electrode materials, various approaches have been undertaken, such as functional group modification, tuning aromaticity or conjugation systems, and polymerization [11,31–33]. Particularly, the utilization of the electron-withdrawing group to control the redox potential is a promising strategy [34–37]. For example, Liang *et al.* demonstrated that the

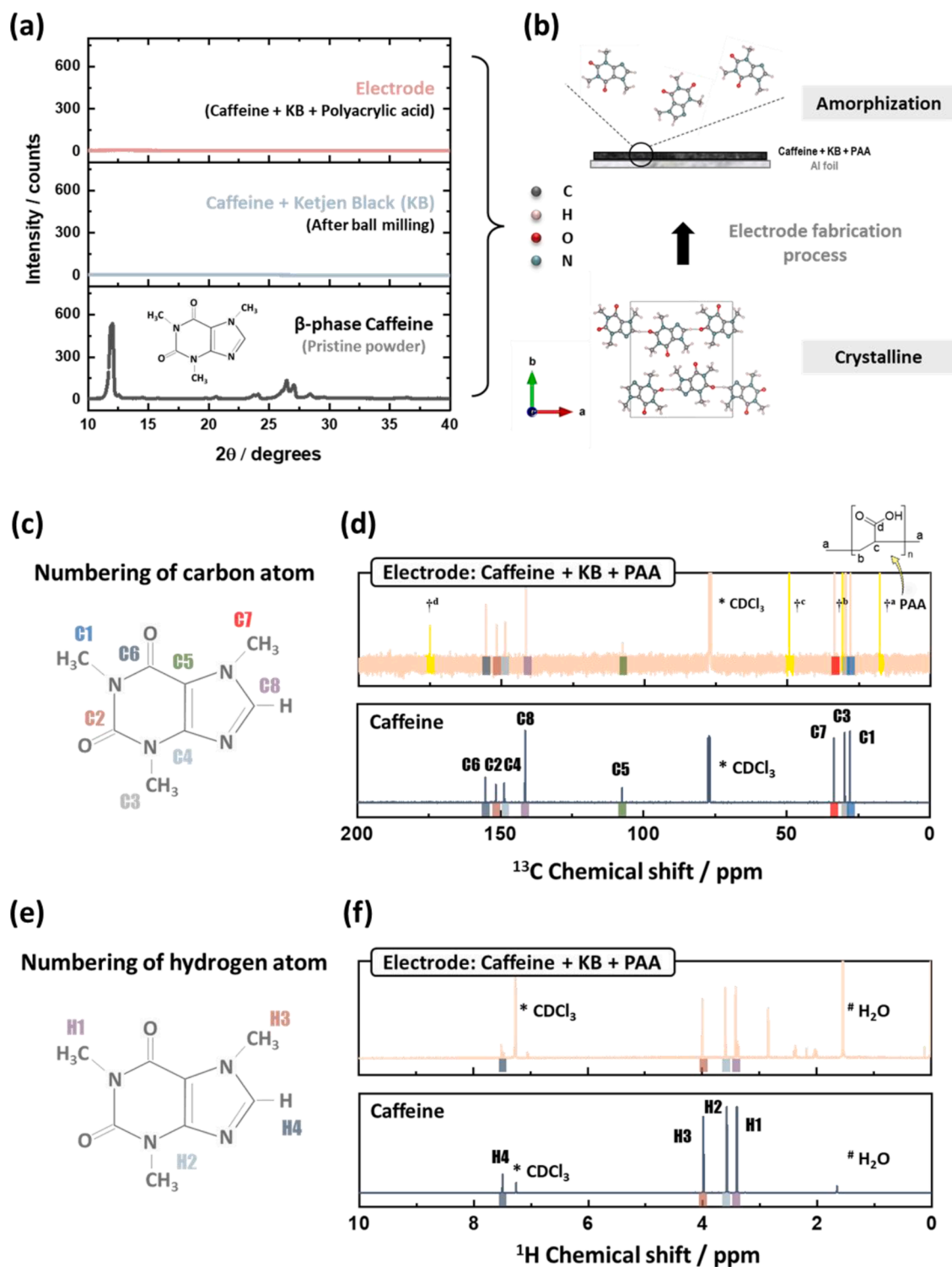


Fig. 1. (a) Observed X-ray diffraction (XRD) patterns ($\lambda = 1.54 \text{ \AA}$) of caffeine powder (β -phase), ball-milled mixture of caffeine and Ketjen Black (KB), and electrode. (b) Schematic illustration of the changes in the crystallinity of caffeine during the electrode fabrication process. Molecular structure and the numbering of (c) carbon and (e) hydrogen atoms in the caffeine compound. (d) ^{13}C and (f) ^1H nuclear magnetic resonance (NMR) spectra of the prepared electrode and caffeine powder, respectively.

working voltage can be modified through the incorporation of fused heteroaromatic by using N, O, and S atoms [35]. In their study, the first reduction potential increases in the order of S (2.52 V), O (2.61 V), and N (2.71 V) electron-withdrawing groups, and these are higher than that of 9,10-anthraquinone (2.27 V). Furthermore, in addition to these potential tuning, finding strategies to activate new redox sites by electron-withdrawing groups could expand the limits of material design for organic electrode energy storage systems.

Alkaloids are natural products or secondary metabolites from various biological groups, including bacteria, fungi, plants, and animals [38]. Alkaloids have a nitrogen atom in their molecular structure, and they are typically used as medicines owing to their pharmacological effects [39]. Caffeine (1,3,7-trimethylpurine-2,6-dione) is a derivative of the xanthine alkaloid, which is the most consumed psychotropic drug [40, 41]. Aside from its stimulant effect, caffeine intake has been reported to lower blood lithium levels [42]. This decrease may be because of

metabolic functions through increased urine production, or there may be a chemical reaction between caffeine and lithium. If caffeine can take up and release lithium-coupled electrons through a redox reaction, it can be used as an energy storage material.

Herein, we demonstrate the feasibility of using caffeine as an electrode material for LIB cathodes and elucidate the energy storage mechanism during the electrochemical reaction. Moreover, a series of halogenated caffeine compounds such as 8-fluoro-1,3,7-trimethylpurine-2,6-dione (8-Fluoro-Caffeine or 8-F-Caffeine), 8-chloro-1,3,7-trimethylpurine-2,6-dione (8-Chloro-Caffeine or 8-Cl-Caffeine), 8-bromo-1,3,7-trimethylpurine-2,6-dione (8-Bromo-Caffeine or 8-Br-Caffeine), and 8-iodo-1,3,7-trimethylpurine-2,6-dione (8-Iodo-Caffeine or 8-I-Caffeine), were synthesized to find a strategy that can modify the reaction voltage or activate new redox sites of caffeine.

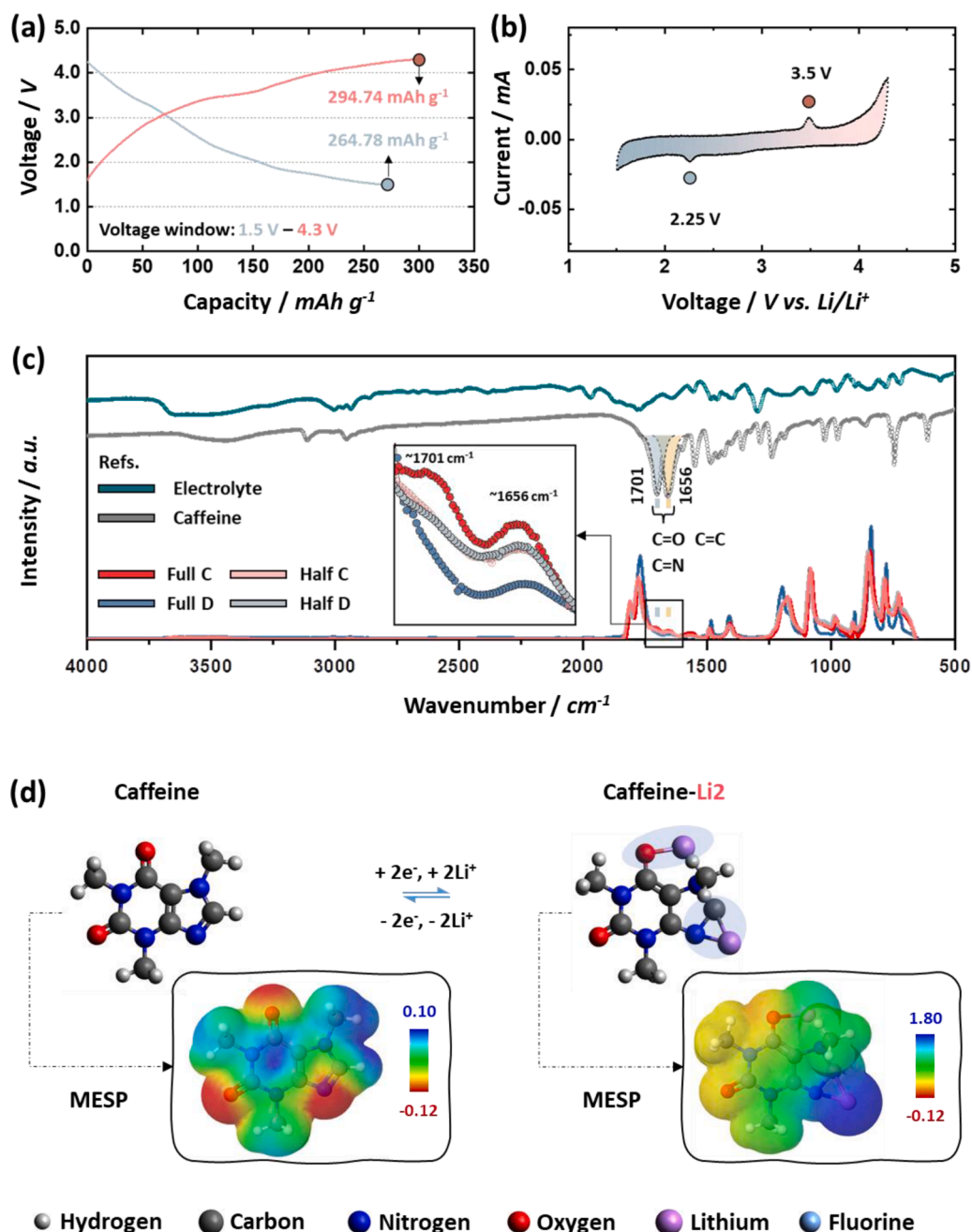


Fig. 2. (a) Voltage profiles of caffeine during the charging and discharging process at a current rate of 0.1C within a voltage range of 1.5–4.3 V. (b) Cyclic voltammograms (potential vs. Li/Li⁺) of caffeine at a scan rate of 0.05 mV s⁻¹ within a voltage range of 1.5–4.3 V. (c) IR spectra of the caffeine electrode at charge, half-charge, discharge, and half-discharge states. (d) Calculated reaction mechanism of pure caffeine and corresponding molecular electrostatic potential (MESP) mapping.

2. Result and discussion

Fig. 1a shows the X-ray diffraction (XRD) pattern of pure caffeine powder and the changes in the XRD patterns during the electrode fabrication process. The observed XRD patterns indicate that the crystal structure of caffeine used in this study is the β -phase, as shown at the bottom of Fig. 1a. However, the peaks of the XRD patterns for the mixture of caffeine and Ketjen Black (KB) after ball milling (middle of Fig. 1a) and for the electrode (top of Fig. 1a) disappeared, indicating that the crystalline material was converted into an amorphous phase during the electrode fabrication process (Fig. 1a,b). This amorphization is considered to be the result of mechanical milling of caffeine and KB during the electrode fabrication process. It is necessary to verify whether caffeine molecules are not decomposed during the electrode fabrication process and exist in the electrode with only loss of crystallinity. The ^{13}C (125 MHz) and ^1H nuclear magnetic resonance (NMR) (500 MHz) spectra of the fabricated electrode and caffeine powder are shown in Fig. 1c–f. Caffeine has eight carbon atoms and four hydrogen atom groups in different environments, and it can be labeled as shown in Fig. 1c,e. In the case of the high-field spectrum for ^{13}C and ^1H NMR, the carbon and hydrogen peaks in the caffeine molecule structure can be easily identified because each chemically inequivalent carbon and hydrogen atom induces a single peak [43,44]. As shown in Fig. 1d,f, the pure caffeine clearly exhibits the 8 and 4 peaks corresponding to the carbon and hydrogen atoms, and the peaks from C1 to C8 and H1 to H4 can be assigned based on the scheme in Fig. 1c,e. Even after the electrode fabrication process, the peak corresponding to caffeine is still present, as shown in Fig. 1d,f, indicating that caffeine is present in the electrode with only loss of crystallinity. In addition, four new peaks (indexed by a, b, c, and d, yellow-colored lines) were observed, which come from the carbon atoms in PAA; the new peak near the peak of C3 is the peak of the "b" carbon atom in PAA.

Fig. 2a shows the voltage profile of the caffeine during the initial cycle, and the charge and discharge capacities are 294.74 and 264.78 mAh g^{-1} , respectively. In the cyclic voltammograms of caffeine, there are two prominent redox peaks at 2.25 V for the cathodic process and 3.5 V for the anodic process (Fig. 2b). The CV graphs from the caffeine electrodes showed a large capacitive current due to the high amount of KB, leading to hard recognition of redox peaks of the caffeine in the graphs. Thus, to help distinguish the redox peak of caffeine, a KB electrode without caffeine was prepared and CV was measured (Fig. S1). To investigate the detailed reaction mechanism of caffeine during the electrochemical reaction, NMR experiments were performed on the electrodes in the charge and discharge states (Fig. S2). First, in the ^1H NMR spectra, it was confirmed that the signals corresponding to the four hydrogen atoms (H1, H2, H3, and H4) were maintained regardless of the charge and discharge states. This indicates that the C–H bond in the caffeine molecule was not broken during the electrochemical reaction, which means that the carbon atoms connected thereto were unlikely to become electrochemically active sites. To find other electrochemically active sites, ^{13}C NMR spectra were obtained, but it was difficult to confirm the carbon peak corresponding to caffeine in both the charge and discharge states. The experiment was performed several times, similar results were obtained. The reason for the overall intensity decrease in the ^1H and ^{13}C NMR spectra is unknown in this study, but it is inferred that the reason only ^1H NMR can be observed is because of the ratio of each atom in nature. The ^{12}C isotope cannot be detected by NMR as it is not magnetically active because its spin quantum number is zero. The ^{13}C isotope, with a spin quantum number of 1/2, can be detected by NMR. However, it exists in nature at 1.1% natural abundance, so the overall sensitivity of ^{13}C NMR is significantly lower relative to that of ^1H NMR [45]. As an alternative, infrared (IR) experiments were performed to find the electrochemically active sites of caffeine. Fig. 2c shows the IR spectra of the caffeine electrodes in the half-discharge, full discharge, half-charge, and full charge states. Pure caffeine powder and electrolyte spectra are also included as references. First, in the case of pure caffeine

(gray-colored spectrum), the peaks at ~ 1656 (yellowish) and ~ 1701 cm^{-1} (blueish) in the IR spectrum correspond to vibrations of the C=O, C=C, and C=N bonds in the caffeine ring [46]. As shown in the inset of Fig. 2c, the intensities of the bands at ~ 1656 and ~ 1701 cm^{-1} decrease in the order of half-to-full discharge during the lithiation process. The reduced peak intensity recovers during the delithiation process by increasing again in the order of half-to-full charge. This indicates that the C=O, C=C, and C=N bonds reversibly participate in the charge compensation reaction during the charge-discharge process.

To make accurate predictions about the Li storage reaction mechanism of pure caffeine, we performed theoretical calculations using density functional theory (DFT), as shown in Figs. S3 and 2d. In the case of pure caffeine, there are three most plausible locations of the inserted Li-ion, i.e., O1-Li & C8-Li-N3 (Fig. S3a), O1-Li & O2-Li (Fig. S3b), and O2-Li & C8-Li-N3 (Fig. S3c). The formation energies are $\Delta G_{(a)} = -6.4661$ eV, $\Delta G_{(b)} = -5.5333$ eV, and $\Delta G_{(c)} = -5.8807$ eV in order of Fig. S3a–c, and O1-Li & C8-Li-N3 show the most energetically stable configuration in the two-electron reactions. Interestingly, a significant decrease in the Gibbs free energy was observed, which seems to be related to structural changes, such as excessive bond angle change in pure caffeine-Li2. (e.g., the stored Li-ion in O1-Li heads on to N element to stabilize the structure). Accordingly, the reaction voltage obtained from the Gibbs free energy calculated by DFT seems to be different from the actual reaction potential. After finding the global minimum of the lithiated caffeine, we inferred the reaction mechanism by utilizing molecular electrostatic potential (MESP) [19,47]. The most probable lithiation positions of caffeine molecule were clearly identified as O1, O2, and N3 atoms through the MESP mapping (Fig. 2d). The emphasis of this result is that pure caffeine reacts with 2 mol of Li-ions and then becomes electrochemically inactive. Considering that the theoretical capacity value of caffeine expressed per lithium or electron is 138 mAh g^{-1} , it is deduced that the C6=O and C8=N bonds act as the main redox-active center during the electrochemical reaction; the initial reversible capacity of caffeine is similar to the theoretical value. Thus, it is conjectured that the decrease in the intensity at ~ 1656 and ~ 1701 cm^{-1} in the IR spectrum in the discharge state is attributed to the loss of vibration of C=O and C=N bonds in caffeine.

Moreover, *ex-situ* XPS analysis was carried out as cross-validation in terms of experimental evidence. As can be seen in Fig. 3a, there are four different components (i, ii, iii, and iv) originating from the sum of the signals of 8 carbon atoms in the caffeine molecule. For nitrogen, the spectrum is made up of four components, corresponding to the four nitrogen atoms in the molecule such as N1, N2, N3, and N4, respectively. However, in the case of oxygen, the individual signals of the two oxygen atoms were not separated in the spectrum, thus only one component was assigned [48]. As predicted from the DFT calculation, the two equivalents of Li ions are stored in the pure caffeine by forming C6-O-Li and N3-Li-C8 bonds at the end of discharge. In Fig. 3b, the relative intensity of C6 signal (iii) decreases, and the peak position shifts to lower binding energy after discharge, suggesting that the original local environment of C6 changed and the electron density of C6 atoms increased. In the case of O 1 s at the end of discharge (Fig. 3d), the relative peak intensity corresponding to the C=O double bond decreases while the intensity of C–O single bond increases (The originally present C–O peak intensity in the O 1 s XPS spectra is inferred to be derived from the C–O bond of the PAA). Therefore, this result demonstrates that the C6=O bond is broken and forms the C6-O-Li bond during the lithiation process, which is predicted by the DFT calculation. As shown in Fig. 3c, the relative intensity of N3 and N4 decreases while the new peak which is inferred as the signal of oxidized N appears near 403.2 eV [49]. At the same time, the ii peak positions (corresponding to signals from C4 and C8) shift to lower energies. This suggests that the N3-Li-C8 bond forms through the oxidation of the N3 atom and the increase in the electron density of C4 and C8.

Fig. 4 shows electrochemical performances, such as cycle and rate capabilities. First, in the case of cycle life characteristics, the discharge

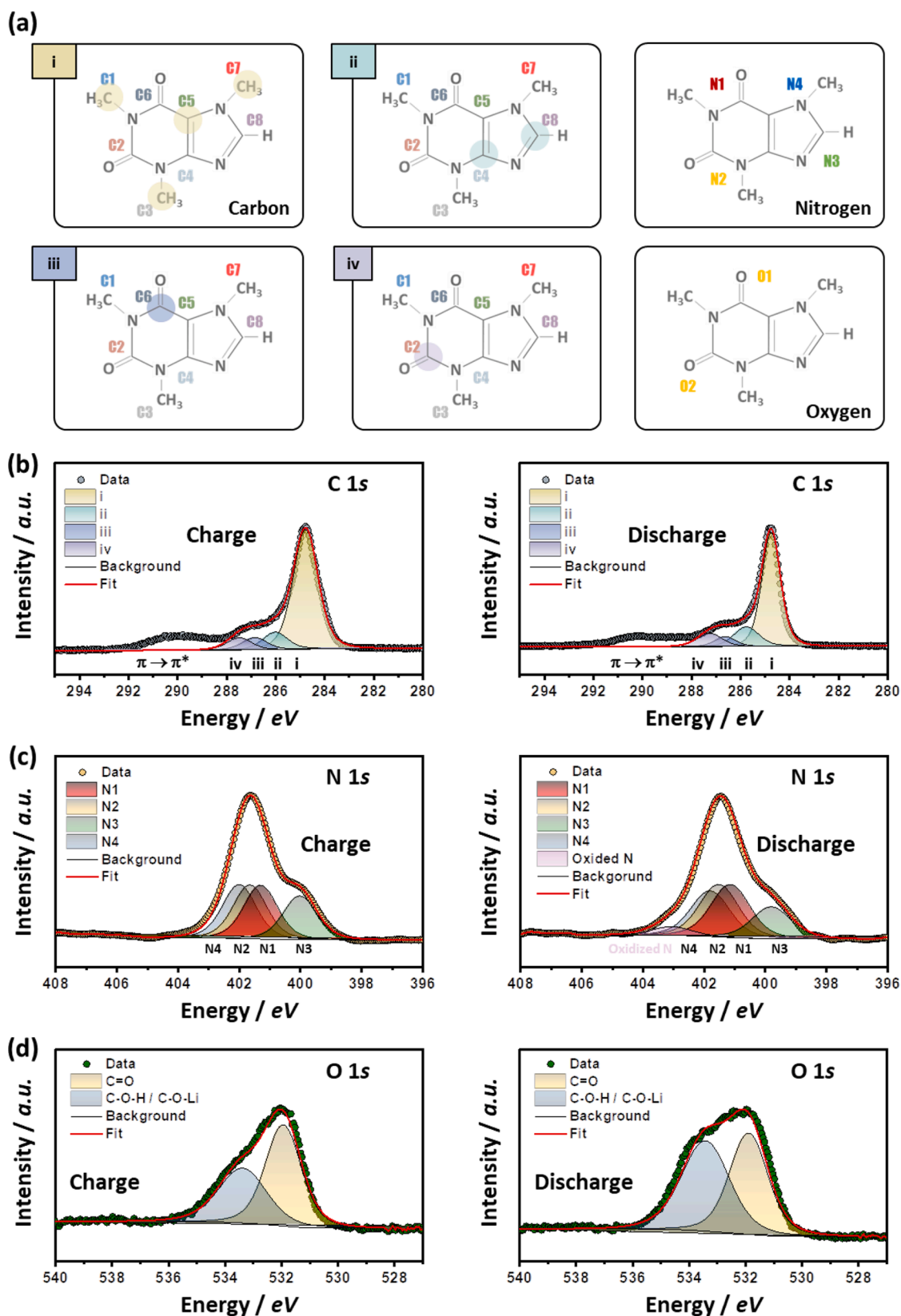


Fig. 3. (a) The carbon group i, ii, iii, and iv is indicated by each different colored circle. Nitrogen atoms are assigned by N1, N2, N3, and N4. For oxygen, there are two assignments such as O1 and O2. (b-d) Ex-situ XPS for the pure caffeine electrode at different states of charge. (b) C 1s core-level XPS. (c) N 1s core-level XPS. (d) O 1s core-level XPS.

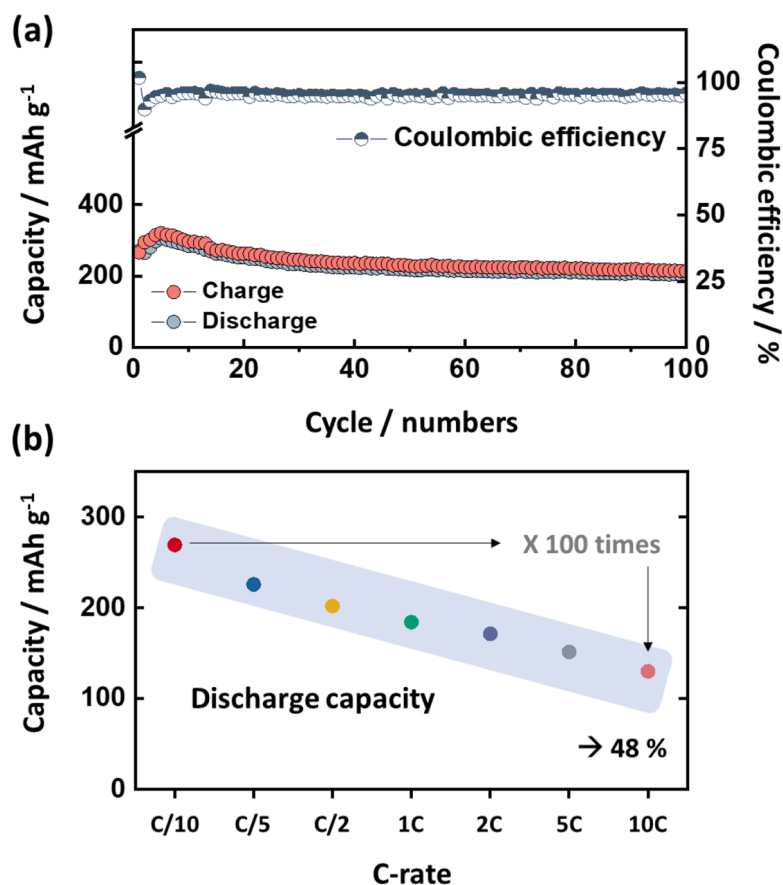


Fig. 4. (a) Cycle performance and (b) rate capability of caffeine material. The cutoff voltage range is 1.5–4.3 V.

capacity decreases from 264.78 to 204.69 mAh g^{-1} after 100 cycles, exhibiting a 77% capacity retention (Fig. 4a). However, there is one thing to note here; the increase in capacity during the initial cycle. As shown in Fig. 2a, the initial charge capacity is $\sim 18.7 \text{ mAh g}^{-1}$ more than the theoretical 276 mAh g^{-1} . Moreover, the capacities of charge and discharge increase during the initial cycles as shown in Fig. 4a. This phenomenon of increasing capacity during cycling, called "negative fading," is sometimes observed in nanoscale engineered transition metal compounds [50]. There are three possible causes of this phenomenon: enhanced re-oxidation, optimization of surface layer, and changes in morphology. However, in the case of organic materials such as caffeine, it is presumed that additional capacity was expressed due to the decomposition reaction of some caffeine molecules during a high voltage charging process such as 4.3 V. This seems to be why caffeine ^1H NMR spectra show a shift (Fig. S2). Afterward, a typical capacity degradation appears, which seems to originate from the general cause of the capacity decrease in organic materials, such as the dissolution of caffeine molecules in the liquid organic electrolyte. For the rate performance, comparing the discharge capacity at 10C with that of C/10, the capacity retention was 48%, even though the rate was approximately 100 times higher (Fig. 4b). This indicates that the energy storage device can be half-charged in only 6 min. However, in this study, the ratio of the active material is low from a commercial point of view because the electrode is constructed in the optimal ratio for the material to express the best capacity in order to identify the reaction mechanism of the material. The excellent electrochemical performances could be achieved with the aid of an excess amount of conductive agent. Therefore, in addition to showing the attractive results used in the study of reaction mechanisms, performance evaluation data of electrodes with an increased ratio of active material are needed to show the disadvantages of caffeine and to signpost for future research directions from a practical

point of view. The electrochemical performances of caffeine electrode material containing 60 wt% of active material are shown in Fig. S4. The charge and discharge capacity has been significantly reduced to $\sim 100 \text{ mAh g}^{-1}$ and the capacity retention is 55.3% after 100 cycles. For rate-capability, caffeine electrodes did not exhibit capacity at high current rates above 1C rate. This means that the caffeine material does not have sufficient conductivity for practical use when the content of the active material in the electrode composite is high. Therefore, in order to be practically used, such a conductivity problem must be solved.

An advantage of organic materials is that electrochemical redox reactions can be modified by substituting atoms in the molecule. To determine whether the electrochemical redox reaction could be modified by intramolecular tuning, the functional groups of the caffeine molecule were engineered via halogenation. Here, H4 attached to C8 was replaced with halogen elements, such as -F, -Cl, -Br, and -I (Fig. 5a). More details on halogenation are provided in the experimental section. Fig. 5b-f show ^1H NMR spectra of the halogenated caffeine molecules. Compared to pure caffeine (Fig. 5b), the signal of H4 at $\sim 7.5 \text{ ppm}$ disappears after halogenation (Fig. 5c-f), indicating that H4 is substituted by halogen elements. Moreover, the peak position of each halogenated caffeine molecule in the region between 3.65 and 3.45 ppm shifts to a lower position in proportion to the electronegativity of the halogen element (Fig. 5g), and halogenated caffeine materials were successfully synthesized. Fig. 5h-l show the cyclic voltammograms of pure caffeine and halogenated caffeine materials. Depending on the substituted halogen element, a unique redox peak appeared in the cyclic voltammogram for each halogenated caffeine material. Although there are several peaks that cannot be explained in this study, one of the most noticeable changes is a new oxidation peak near 4 V. Compared with pure caffeine, all halogenated caffeine substances have a new oxidation peak at approximately 4 V, and the voltages increase sequentially in the

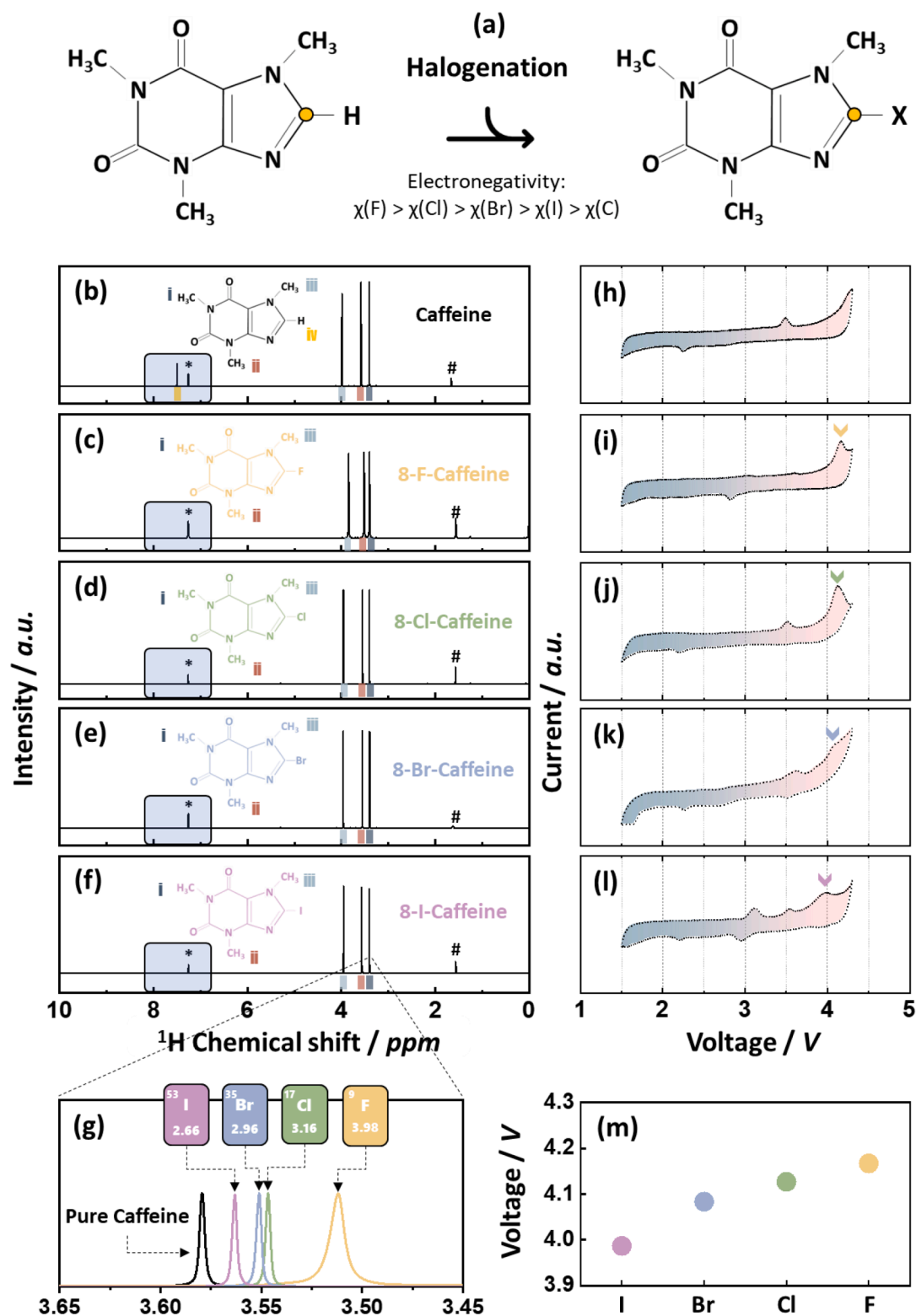


Fig. 5. (a) Schematic illustration of halogenation of caffeine. ¹H NMR spectra for (b) caffeine, (c) 8-F-Caffeine, (d) 8-Cl-Caffeine, (e) 8-Br-Caffeine, and (f) 8-I-Caffeine. (g) Magnified region between 3.65 and 3.45 ppm where all five spectra overlap. Cyclic voltammograms of (h) caffeine, (i) 8-F-Caffeine, (j) 8-Cl-Caffeine, (k) 8-Br-Caffeine, and (l) 8-I-Caffeine at a scan rate of 0.05 mV s⁻¹ in a voltage range of 1.5–4.3 V. (m) Comparison of the new oxidation peaks observed at approximately 4 V in the four halogenated caffeines.

order of electronegativity as -I, -Br, -Cl, and -F as shown in Fig. 5m. To determine why this new oxidation peak occurs and why the reaction potential varies depending on the substituted element, the reaction mechanism for the halogenated caffeine (e.g., 8-F-Caffeine) is predicted

by the DFT calculation.

In the case of F-substituted caffeine (8-F-Caffeine), there are also three most plausible configurations in the two-electron reaction (Fig. S5a-c). Among the three cases, the case of O1-Li & C8-Li-F shows

the lowest formation energy ($\Delta G_{\text{a}} = -9.8772$ eV), which means that the corresponding configuration is likely to occur with the highest probability in a two-electron reaction. In the same way, MESP was used to infer the reaction mechanism of 8-F-Caffeine. Contrary to pure caffeine (Fig. 2d), 8-F-Caffeine still has an active lithiation site after two moles of Li-ion are lithiated, as indicated by the red color of the negative isosurface value (Fig. 6). As a result, the 8-F-Caffeine can store a total of three Li-ions through the C2=O bond breakage, resulting in a larger capacity than pure caffeine. However, the point to note is that the isosurface value of C8-Li-F in MESP is still negative even after lithiation of 3 mol of Li-ion. Thus, it is inferred that this triggers an undesirable reaction during subsequent cycles, resulting in the dissociation of the F ion from the caffeine molecule. This may be why halogenated caffeine loses its high capacity in the early cycles. To confirm this prediction, IR spectra according to the degree of lithiation and *ex-situ* XPS spectra were obtained depending on charge and discharge states.

Fig. 7a exhibits that the intensities of the bands at ~ 1656 and ~ 1701 cm $^{-1}$ gradually decrease as the degree of lithiation increases, and the intensity at the end of the discharge state (*i.e.*, discharged to 1.5 V) is much lower than that of pure caffeine. Therefore, in the case of halogenated caffeine, it can be inferred that two C=O bonds participate in the Li storage reaction, unlike pure caffeine in which one C=O bond is merely active during the lithiation process. In the case of 8-F-Caffeine after discharge, the C 1 s XPS spectra show that the relative peak intensity of C2 and C6 decreases, and their peak position shift to low energy (Fig. 7b). Correlated to this, the relative peak intensity of the C=O bond decreases while that of the C—O bond increases as shown in O 1 s XPS spectra (Fig. 7d). The noticeable thing is that the relative intensity of the C—O bond in the 8-F-Caffeine surpasses that of the C=O bond, compared to the case of the O 1 s XPS spectra in pure caffeine. From the C 1 s and O 1 s XPS spectra, it is deduced that the two equivalents of Li ions are stored in the 8-F-Caffeine through the breakage of C2=O and C6=O bonds. For N 1 s XPS spectra (Fig. 7c), there is no significant change after discharge compared to the case of pure caffeine, which is similar to the prediction in the DFT calculation. Furthermore, it is

inferred that the increasing trend of oxidation potential is owing to the different inductive effects of each halogen element depending on their electronegativity (that is, the electronegativity: $\chi(\text{C}): 2.55 < \chi(\text{I}): 2.66 < \chi(\text{Br}): 2.96 < \chi(\text{Cl}): 3.16 < \chi(\text{F}): 3.98$).

As shown in Fig. 8a, the halogenated caffeine materials show discharge capacities of over 400 mA h g $^{-1}$, which is close to the theoretical value of 414 mA h g $^{-1}$ when 3 mol of Li-ions participate in the charge compensation. Hence, it is evident that halogenation can create a new reaction site for the caffeine molecule, which previously reacted with only 2 mol of lithium. However, all halogenated materials exhibit a severe deterioration in the cycle life during the long electrochemical reaction (Fig. 8b). To get a hint about the cause of this phenomenon, we measured the ^1H NMR of 8-F-Caffeine with the best cycle stability and 8-I-caffeine with the worst among the substituted materials (Fig. S3). First of all, the most notable thing is that the H4 peak reappears after 5 cycles of both halogenated caffeine materials, as shown in Fig. S6. This indicates that the substituted halogen elements were detached from the caffeine molecule and hydrogen atoms were reattached to C8 site. As a result, the new redox active sites were inactivated, and 8-F-Caffeine exhibited a reduced reversible capacity which is similar to the original caffeine during 100 cycles. This is consistent with the theoretical prediction of the DFT calculation for 8-F-Caffeine. In addition, a new peak also appeared in the region of H1, indicating that a new hydrogen environment formed in the caffeine molecule. Moreover, even the peak intensity of the original hydrogen H1 almost disappeared. This suggests that the side reaction involves the release of the alkyl group of the caffeine molecules. Compared to 8-F-Caffeine, 8-I-Caffeine showed more new hydrogen peaks in the ^1H NMR spectrum. This indicates that the 8-I-Caffeine electrode material contains more newly formed local hydrogen environment than the 8-F-Caffeine through side reactions, and the newly formed local environment causes abnormal electrochemical reactions.

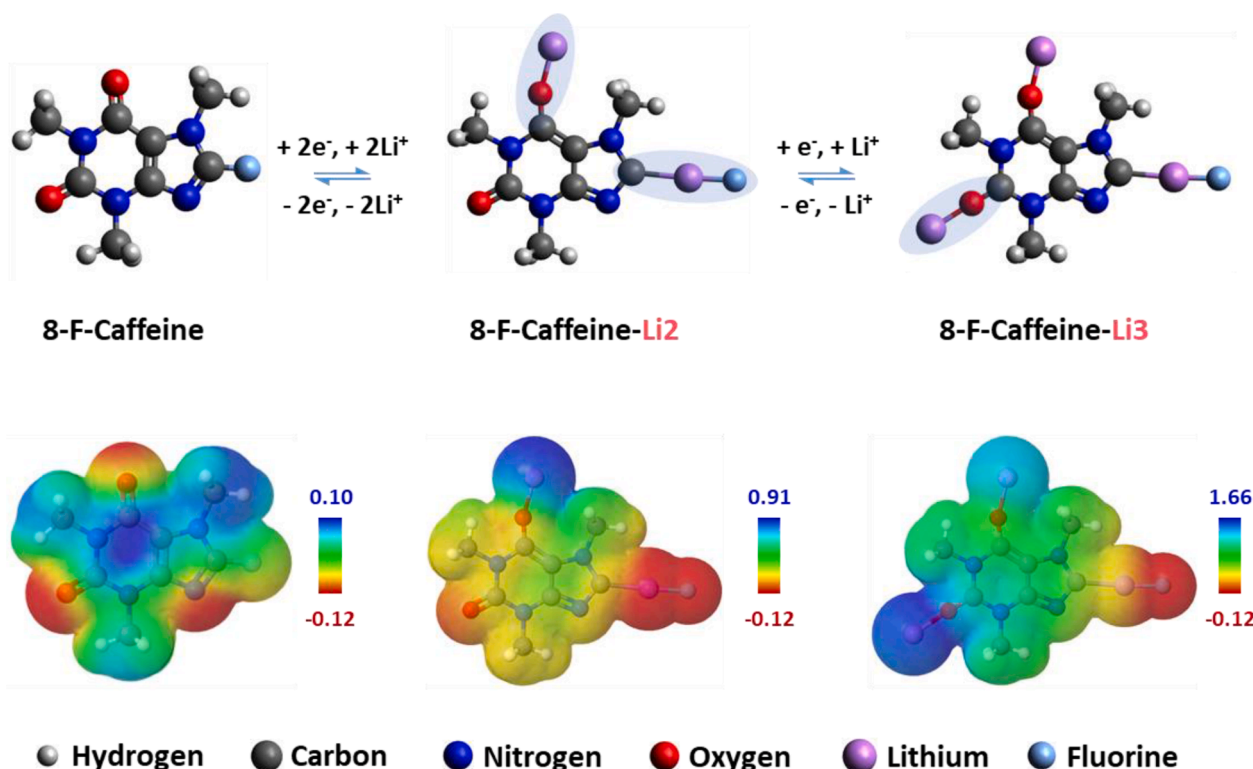


Fig. 6. Calculated reaction mechanism of F-substituted caffeine (8-F-Caffeine) and corresponding molecular electrostatic potential (MESP) mapping.

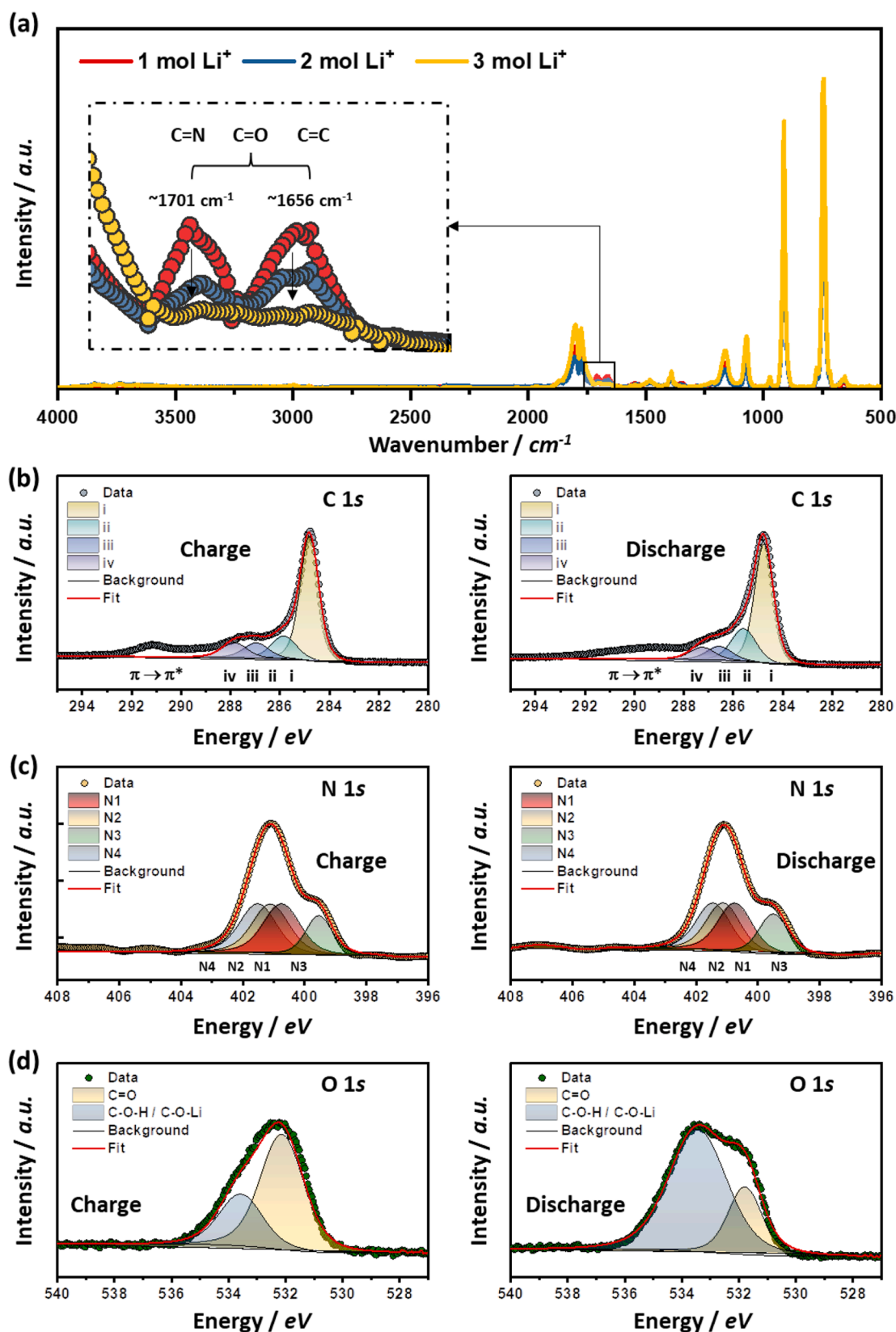


Fig. 7. (a) IR spectra of the 8-F-Caffeine electrode at the state of 1 mol, 2 mol, and 3 mol lithiation. *Ex-situ* XPS for the halogenated caffeine electrode (8-F-Caffeine) at different states of charge. (b) C 1 s, (c) N 1 s, and (d) O 1 s core-level XPS.

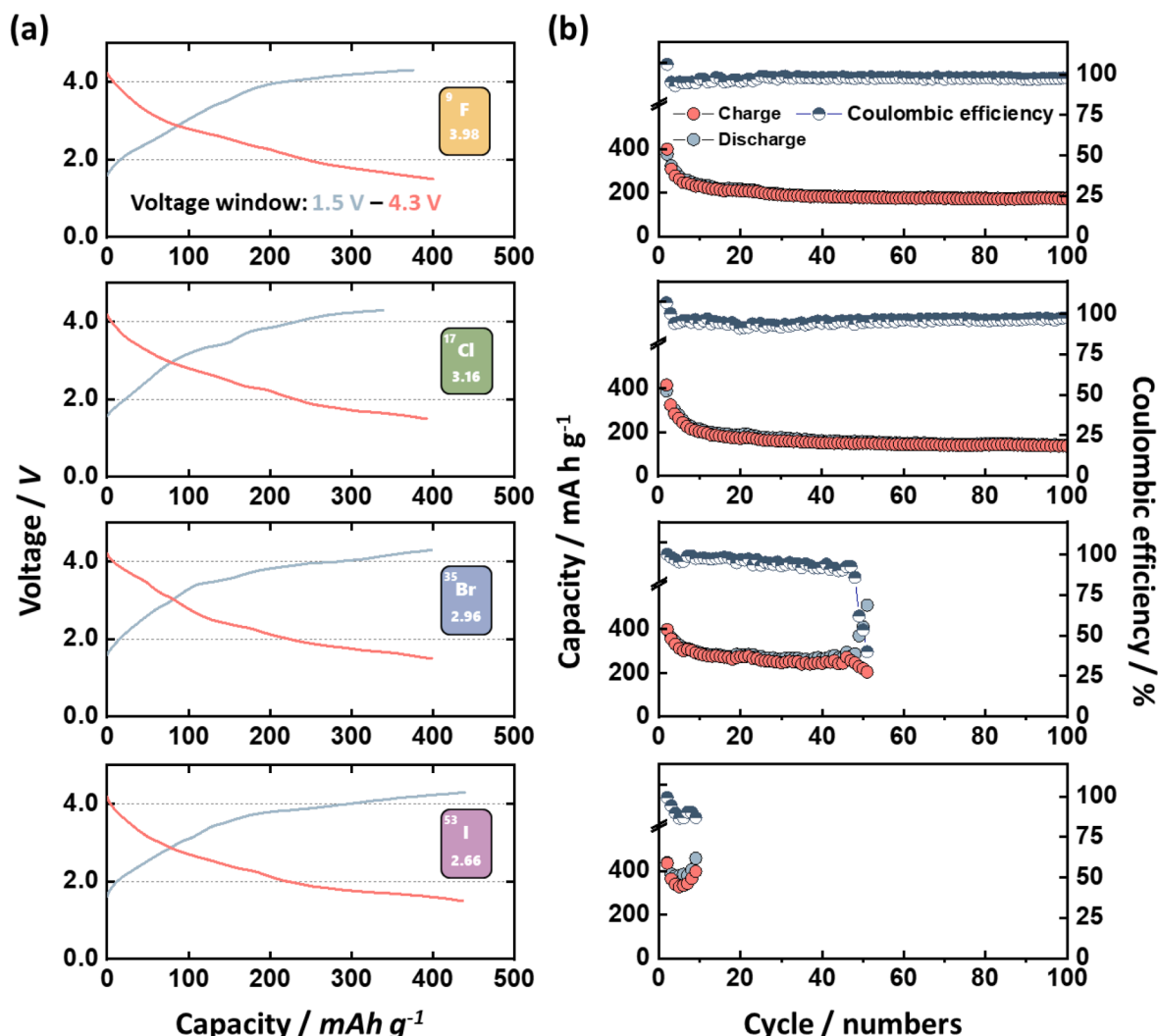


Fig. 8. (a) First charge-discharge curves and (b) cycle performance during 100 cycles of halogenated caffeine material. The cutoff voltage range is 1.5–4.3 V. Current rate: C/10.

3. Conclusion

In this study, we demonstrated the feasibility of caffeine as an energy storage material and elucidated its reaction mechanism. Furthermore, we revealed that modifying the functional groups at the molecular level can create new electrochemical reaction sites. Reversible redox reactions of caffeine were confirmed by galvanostatic charge-discharge, and it exhibited a performance exceeding 200 mAh g^{-1} capacity, even after 100 cycles in the voltage range of 1.5–4.3 V. The energy storage mechanism, in which two equivalents of lithium and electrons react around the carbonyl group (C=O) and imine group (C=N) of the caffeine molecule, was demonstrated through DFT, NMR and IR characterization. Moreover, it has been determined that a new redox site can be introduced into caffeine molecules via halogenation, and the reaction potential can be controlled based on the electronegativity of the substituted functional group by NMR and cyclic voltammetry. Although organic materials have intrinsic challenges such as low redox potential, low electrical conductivity, and high solubility in liquid organic electrolyte systems, their cost-effectiveness, sustainability, and design flexibility make them appealing as energy storage materials. Furthermore, a natural and human-edible compound would be a promising option for use as an energy storage material. However, caffeine and its derivatives still require various optimization studies, such as the alleviation of voltage fading and high redox polarization, as well as the issues

mentioned above. Therefore, further studies such as the fabrication of nano-carbon composites to enhance electrical conductivity and the optimization of electrolytes to improve cycle retention are required. In addition, a clear explanation is required for the detailed reaction mechanism derived from the newly formed reaction sites through molecular remodeling, and the deterioration mechanism. This study suggests that if various naturally derived compounds with less toxic can be utilized as energy storage materials, and their electrochemical properties and performances can be improved by molecular-level modifications, it can provide a variety of options for designing next-generation rechargeable batteries.

CRediT authorship contribution statement

Wontae Lee: Supervision, Project administration, Funding acquisition, Conceptualization, Methodology, Visualization, Investigation, Resources, Writing – original draft, Writing – review & editing. **Yeongjin Lee:** Methodology, Investigation, Validation, Formal analysis, Writing – review & editing. **Hyunyoung Park:** Formal analysis. **Munhyeok Choi:** Investigation, Validation, Formal analysis. **Sungho Kong:** Investigation, Validation, Formal analysis. **Jung Woon Yang:** Supervision, Conceptualization, Resources, Writing – review & editing. **Jongsoon Kim:** Supervision, Formal analysis. **Won-Sub Yoon:** Supervision, Project administration, Conceptualization, Methodology, Resources, Writing –

review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.ensm.2023.01.003.

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